

Department of Information Science & Engineering



“Smart Cultivation and Marketing of Indigenous Fruit Crops”

Minor Project (BIS586)

Submitted by

Suraj M S	4NI23IS217
Shrinivas Mattur	4NI23IS202
Yashwanth H S	4NI23IS248
Yadhunandan G	4NI23IS246

Under the Guidance of

Dr.Nandini B M

Associate Professor

Dept of IS&E

NIE, Mysuru



The National Institute of Engineering
(Autonomous Institution under Visvesvaraya Technological University)
MYSURU – 570018
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The National Institute of Engineering
Manandawadi Road, Mysuru-570008

Department of Information Science & Engineering



CERTIFICATE

Certifies that the Minor-Project titled “**Smart Cultivation and Marketing of Indigenous Fruit Crops**” is presented by **Suraj M S (4NI23IS217), Shrinivas Mattur (4NI23IS202), Yashwanth H S (4NI23IS248), Yadhunandan G (4NI23IS246)** in partial fulfilment for the requirements of the fifth semester BE in Information Science & Engineering prescribed by The National Institute of Engineering, Autonomous Institution under Visvesvaraya Technological University, Belagavi. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated. The Minor- Project report has been approved as it satisfies the academic requirements in respect of the minor project prescribed for the fifth semester.

Signature of Guide

Dr. Nandini B M
Associate Professor
Dept. of ISE
NIE , Mysuru

Signature of HOD

Dr. Girish
HOD & Professor
Dept. of ISE
NIE , Mysuru

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Suraj M S	4NI23IS217
Shrinivas Mattur	4NI23IS202
Yashwanth H S	4NI23IS248
Yadhunandan G	4NI23IS246

ABSTRACT

The project titled “**Smart Cultivation and Marketing of Indigenous Fruit Crops**” aims to assist farmers in cultivating and promoting traditional fruits such as **Ramphala, Lakshmanphala, and Wood Apple** using **Artificial Intelligence (AI)** and **Machine Learning (ML)** technologies. The system is designed to analyze **soil parameters** and **climatic conditions** to recommend the most suitable crops for a particular region. It also integrates a **leaf disease detection module**, which enables farmers to upload images of diseased leaves through a smartphone camera. The AI model processes these images to identify plant diseases at an early stage, helping to minimize yield losses and ensure timely intervention.

Additionally, the system incorporates a **database** to store and retrieve information related to crops, market demand, and farmer details. It further provides **visual marketing insights** by analyzing production and trends, empowering farmers to make informed decisions and increase profitability. By combining AI-driven analysis, data visualization, and sustainable agricultural practices, the project promotes both technological advancement and the economic well-being of farmers

This project bridges the gap between **traditional agricultural knowledge and modern technology**, ensuring that indigenous fruit crops are cultivated sustainably while providing farmers with digital tools for real-time decision-making. By automating crop selection, disease detection, and market forecasting, the system helps **increase yield, reduce losses, and enhance profitability**.

Moreover, the project contributes to **sustainable agricultural development** by promoting biodiversity, reducing dependence on chemical-based farming, and creating awareness of the nutritional and economic importance of native fruits. Through this initiative, farmers can transition from conventional practices to a **data-driven, AI-supported agricultural model** that strengthens food security and rural livelihoods.

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Chapter 01

Introduction

1.1 Problem Statement

In modern agriculture, farmers often face significant challenges in making informed decisions regarding crop selection, disease management, and market accessibility. Traditional methods rely heavily on manual observation, local experience, and guesswork, which are not always accurate or efficient. As a result, farmers encounter frequent yield losses, poor soil utilization, and limited profit margins.

Additionally, the cultivation of indigenous fruit crops such as Ramphal, Lakshmanaphal, and Wood Apple has declined due to lack of awareness, inadequate technical guidance, and absence of digital tools that support their growth and marketing. Disease outbreaks frequently go undetected until they cause severe damage, and farmers lack real-time access to scientific or market data.

1.2 Objectives

- To assist farmers in sustainable cultivation of indigenous fruit crops such as Ramphal, Lakshmanaphal, and Wood Apple through AI-based guidance.
- To recommend suitable crops by analyzing soil properties — pH, moisture, and nutrient composition — using machine-learning algorithms.
- To detect plant diseases automatically from uploaded leaf images using deep-learning models and provide early warnings.
- To develop a data-driven marketing platform that visualizes market trends, demand, and pricing for indigenous fruits.
- To increase farmer profitability and awareness about the nutritional and commercial potential of native fruit crops.

1.3 Purpose

- The purpose of the project is to bridge the gap between traditional farming practices and modern data- driven agriculture.
- Farmers often lack access to scientific tools that analyze soil quality or identify diseases accurately. This system leverages Artificial Intelligence and Machine Learning to make agriculture smarter, faster, reliable.
- It empowers farmers to take timely, informed decisions on crop selection, fertilizer usage, and disease control, while also helping them connect with larger markets and consumers. Ultimately, the purpose is to enhance productivity, ensure sustainability, and preserve indigenous biodiversity.

1.4 Existing System

- Crop selection is based mainly on experience or trial-and-error, not scientific data.
- Disease identification is done manually, often too late to prevent crop loss.
- Farmers rely on local traders or intermediaries for selling produce, limiting their profit margins.
- There is no unified digital platform that integrates soil analysis, disease detection, and marketing for indigenous crops.

These limitations lead to reduced yield, wastage, and economic instability among small-scale farmers.

1.5 Proposed System

The proposed Smart Cultivation and Marketing System introduces an AI-powered, integrated platform that supports farmers throughout the cultivation cycle.

Key features include:

- **AI-based Crop Recommendation:** Uses soil parameters to predict the most suitable fruit crops.
- **Deep-Learning Disease Detection:** Processes leaf images using CNN models to identify diseases accurately.
- **data-driven platform** to visualize market trends, demand, and pricing of indigenous fruits. It aims to enhance farmer profitability and awareness of the nutritional and commercial value of native fruit crops.
- **User-Friendly Web Interface:** Allows farmers to upload data, receive instant feedback, and access marketing information easily.

The project, "Smart Cultivation and Marketing of Indigenous Fruit Crops," aims to leverage Artificial Intelligence (AI) and Machine Learning (ML) to enhance the cultivation and market profitability of valuable indigenous Indian fruits, specifically Ramphala (Custard Apple), Lakshmanphala (*Baccaurea ramiflora*), and Wood Apple (*Limonia acidissima*)[15].

1.6 The Potential of Indigenous Fruit Crops:

- **Ramphala** (*Manilkara hexandra*): also shows antioxidant activity, which helps protect the body from oxidative stress and cellular damage[1]. Its bark extracts exhibit mild antimicrobial effects, making it useful in managing minor infections. Additionally, the plant is sometimes used in traditional remedies to support liver function and overall digestive health, enhancing its value in herbal medicine[3].



Figure 1.1 Ramphala

- **Lakshmanphala** (*Baccaurea ramiflora*): is a nutrient-rich fruit known for its potent **antioxidant properties**, which help neutralize free radicals and protect cells from damage. It contains bioactive compounds such as **acetogenins**, which have been studied for their **anticancer potential** due to their ability to inhibit abnormal cell growth. Traditionally in Ayurveda, it is used to treat **skin diseases**, improve complexion, and support **liver health** by aiding detoxification[2]. Additionally, the fruit shows **anti-inflammatory and antimicrobial activities**, making it beneficial for managing minor infections and inflammatory conditions. Its nutrient profile also supports **digestive wellness**, and some folk practices use it to boost overall immunity and vitality[1].



Figure 1.2: Lakshmanphala

- **Wood Apple** (*Limonia acidissima*): is often regarded as a ‘**Superfruit**’ due to its dense nutrient profile, especially its high levels of **Vitamin C, Calcium, and Potassium**, which support immunity, bone strength, and electrolyte balance. In traditional medicine, the pulp is widely used to manage **digestive problems** such as diarrhea, dysentery, and constipation because of its natural astringent and laxative properties[5][6].



Figure 1.3 :Wood Apple

Chapter 02

LITERATURE SURVEY

[1] Paper: “*AI-Powered Crop Recommendation and Soil Analysis Using Machine Learning*”

by K. Patel, R. Mehta, and S. Sharma (2024)

This paper presents a machine learning–based crop recommendation system that analyzes soil characteristics such as pH, nitrogen, phosphorus, potassium (NPK), and moisture levels. The authors used supervised learning algorithms like Random Forest and Decision Tree to predict the best-suited crop for a given soil profile. The study emphasizes the role of data-driven decision-making in improving crop yield and optimizing fertilizer use. The methodology outlined in this paper provides foundational insights for implementing the soil-based crop recommendation module in our project.

[2] Paper: “*Plant Disease Detection Using Deep Learning and Image Processing*” by A. Verma and N. Raj (2021)

This research focuses on using Convolutional Neural Networks (CNNs) and image preprocessing techniques for early detection of plant diseases through leaf images. The model achieves high accuracy by employing image augmentation, segmentation, and feature extraction methods. The paper highlights how CNNs outperform traditional image classification methods in identifying plant infections[12]. This study forms the basis for our project’s AI-based disease detection module, enabling farmers to detect and treat diseases promptly.

[3] Paper: “*AI in Agriculture: A Review of Applications and Challenges*” by R. Singh, M. Kumar, and T. Das (2025)

This review paper provides an overview of how artificial intelligence is revolutionizing modern agriculture through applications in crop monitoring, irrigation control, and predictive analytics. It discusses challenges such as data scarcity, model generalization, and farmer adoption barriers. The findings emphasize the need for user-friendly AI systems that can be easily adopted by small and medium-scale farmers. These insights are valuable for designing an intuitive AI-powered farming assistant as proposed in our system.

**[4] Paper: “*Market Forecasting and Price Prediction in Agriculture Using Machine Learning*”
by S. Reddy and P. Srinivasan (2022)**

This paper explores the use of machine learning algorithms for predicting agricultural commodity prices using historical market data and weather patterns. Models such as LSTM (Long Short-Term Memory) and Linear Regression were used to forecast price trends, assisting farmers in making informed selling decisions. The research demonstrates that integrating price prediction tools within agricultural systems can enhance farmers’ income. This work supports the marketing and visualization component of our project, which aims to give farmers access to real-time market insights.

**[5] Paper: “*Sustainable Cultivation and Conservation of Indigenous Fruit Crops in India*” by
Dr. M. Prakash and V. Iyer**

This paper emphasizes the importance of conserving indigenous fruit species such as Ramphal, Lakshmanphala, and Wood Apple, which possess unique nutritional and medicinal properties. The study discusses challenges faced by farmers in propagating and commercializing these crops, including limited awareness and market reach. The authors advocate for the integration of modern technologies like IoT and AI to promote sustainable cultivation practices. This work directly aligns with our project’s objective of enhancing indigenous crop cultivation and marketing through technology.

CHAPTER 3:

SYSTEM ANALYSIS

System analysis involves understanding requirements and designing a logical structure for the Smart Cultivation System to meet farmer needs efficiently.

3.1 Advantages and Disadvantages

Feature	Advantage (Benefits)	Disadvantage (Challenges)
Crop Suggestion	Increases yield and resource efficiency by matching crop to soil.	Model accuracy is dependent on the quality and comprehensiveness of the training dataset.
Disease Detection	Enables early, non-expert diagnosis, preventing widespread crop loss and ensuring timely treatment.	Requires good quality images and a robust, trained deep learning model; high GPU power may be needed for training.
Marketing	Increases farmer income by promoting high-value indigenous crops and connecting farmers to e-commerce/markets.	Requires continuous updating of market trend data and effective community outreach.
Indigenous Focus	Promotes sustainable cultivation of valuable, nutrient-rich plants.	The fruits (Ramphala, Lakshmanphala, Wood Apple) may have a limited, specialized market compared to staple crops.

3.2 Cost Effectiveness

The solution is expected to be highly cost-effective for farmers in the long run:

- **Reduced Input Costs:** Accurate crop suggestions based on soil analysis minimize wasteful expenditure on unsuitable seeds, fertilizers, and unnecessary resources.
- **Lower Loss due to Disease:** The AI-powered early disease detection significantly reduces the financial losses caused by crop diseases, minimizing the time between symptom appearance and treatment.

- **Increased Profitability:** By shifting focus to high-demand indigenous fruits and enabling effective, data-driven marketing, the project directly aims to increase farmer income.
- **Leveraging Open-Source Technology:** The software uses open-source libraries like **Python**, **scikit-learn**, and **PyTorch**, reducing licensing costs. The main cost is the initial development and the recommended hardware for faster model training.

3.3 Comparison With Existing and Proposed System

Feature	Existing System	Proposed System
Technology Used	Mostly manual methods and basic mobile apps for data entry.	Advanced AI & ML algorithms for crop recommendation, disease detection, and market analysis.
Crop Recommendation	Based on farmer's experience and general climate information.	Uses soil parameters (pH, NPK, moisture) analyzed through Machine Learning models for accurate suggestions.
Disease Detection	Manual visual inspection; errors are common and slow to detect.	Automated AI-based leaf image analysis (CNN) for early and accurate identification of plant diseases.
Data Storage	Records maintained manually or through spreadsheets.	Centralized Database for secure and organized storage of crop, farmer, and market data.
Marketing Support	Farmers depend on local traders and middlemen.	Provides data visualization and digital market insights , connecting farmers directly with buyers and online platforms.
Sustainability	Focused mainly on commercial hybrid crops.	Encourages cultivation of indigenous fruit crops for biodiversity and ecological balance.
Decision-Making	Experience-based, often imprecise.	Data-driven and automated , improving productivity and profitability.

3.4 System Requirements

Software Requirements :

- **Operating System:** Windows 10/11 or Linux
- **Programming Language:** Python 3.8+
- **Database:** MySQL or SQLite
- **Libraries:**
 - **Data Handling & Analysis:** Pandas, NumPy
 - **Machine Learning (Crop Suggestion):** Scikit-learn (for Random Forest)
 - **Deep Learning (Disease Detection):** PyTorch, Torchvision, TensorFlow / Keras[9] (alternative), OpenCV[14].
 - **Visualization (Marketing):** Matplotlib, Seaborn, Plotly, Power BI
- **Tools & IDEs:** VS Code, Streamlit

Hardware Requirements :

- **Processor:** Intel i5 / AMD Ryzen 5 or higher (Recommended for model training)
- **Memory (RAM):** Minimum 8 GB (Recommended: 16 GB)
- **GPU:** NVIDIA GPU (e.g., GTX 1650 or higher) - Optional but Highly Recommended for faster deep learning model training
- **Storage:** 256 GB SSD or more (for fast dataset loading and model storage)
- **Other Devices:** Smartphone/digital camera for leaf image capture, stable internet connection

CHAPTER 4:

SYSTEM DESIGN

4.1 Design Methodology

The project follows a modular design methodology to ensure each core function (Crop Suggestion, Disease Detection, Marketing) can be developed and integrated independently.

- **Data-Driven Design:** The entire system is centered around data: a structured dataset for crop prediction, an image dataset for disease classification, and market data for strategic insights.
- **Modular Architecture:**
 1. **Frontend/User Interface:** Designed for user input (soil parameters) and image upload, accessible via a web or mobile interface (conceptualized for a prototype).
 2. **Business Logic/ML Server:** Hosts the trained ML models (Random Forest for crop suggestion, CNN for disease detection). This server receives the input, processes it using the respective models, and returns the output (suggested crop/disease diagnosis).
 3. **Data Storage :** A backend database stores all static and dynamic project information: fruit details, cultivation practices, disease management, and market data.
 4. **Analytics Layer:** Periodically extracts data from the Data Base to generate market-focused visualizations using **Power BI** or Python libraries, providing actionable insights

4.2 Design Flowchart

This flowchart designed to assist users, primarily farmers, in managing crop cultivation .The system begins with a module selection step offering four main functionalities: providing detailed **Fruits Information** (including cultivation factors and nutritional/economic uses for Custard Apple); accepting **Input Soil Parameters** to run an **ML Model** for **Recommended Fruit** suggestion; allowing users to **Upload a Leaf Image** for **Disease Prediction** and subsequent display of **Symptoms & Remedies**; and viewing **Marketing Insights** related to cultivation areas, commercial data, and health information, relevant to both the farmer and consumer. All data, recommendations, and diagnostic

results from the ML and diagnosis modules are ultimately directed to a central **Visualization Dashboard**, which serves as the comprehensive output interface before the process concludes.

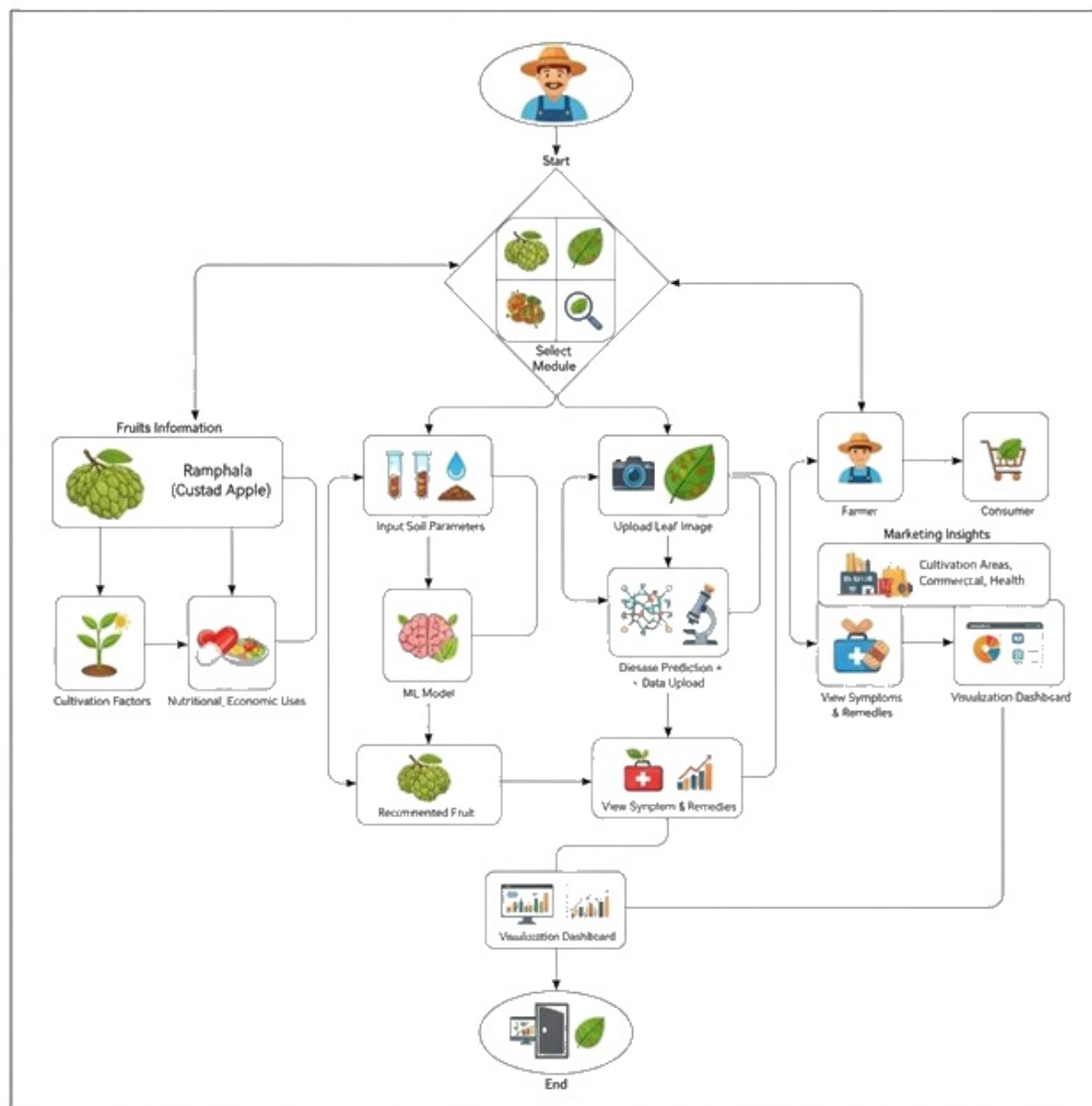


Figure 4.1: Design Flowchart of Smart Cultivation

4.3 Data Flow Diagram (DFD)

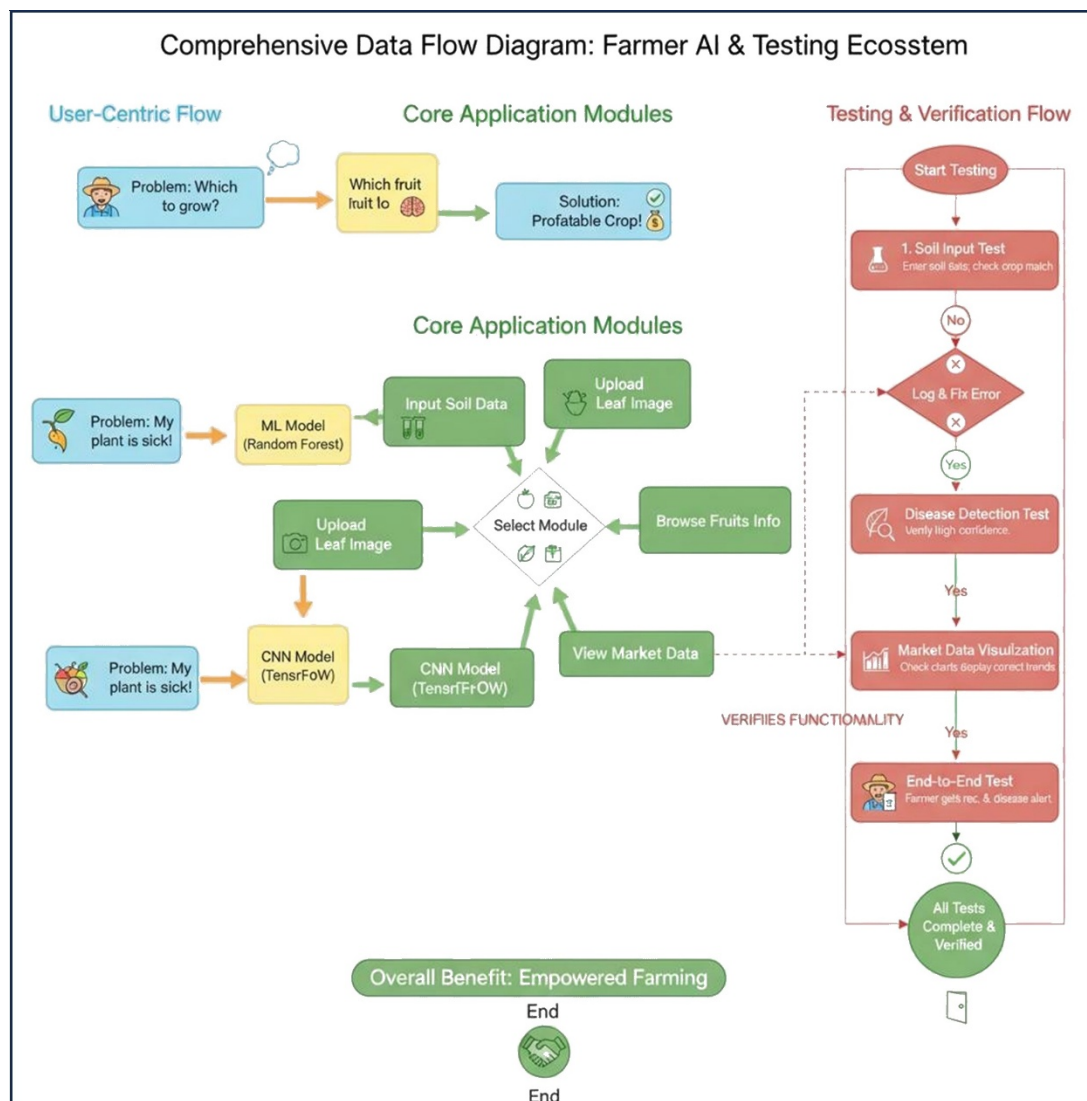


Figure 4.2 :Data Flow diagram

this data flow diagram illustrates a robust AI application for farming that not only provides the user with crop recommendations (via soil data) and disease diagnosis (via leaf images and ML/CNN models) but also integrates a rigorous Testing and Verification Flow to ensure the accuracy and reliability of its core functions before the final end-to-end functionality is delivered to the farmer.

4.4 System Architecture:

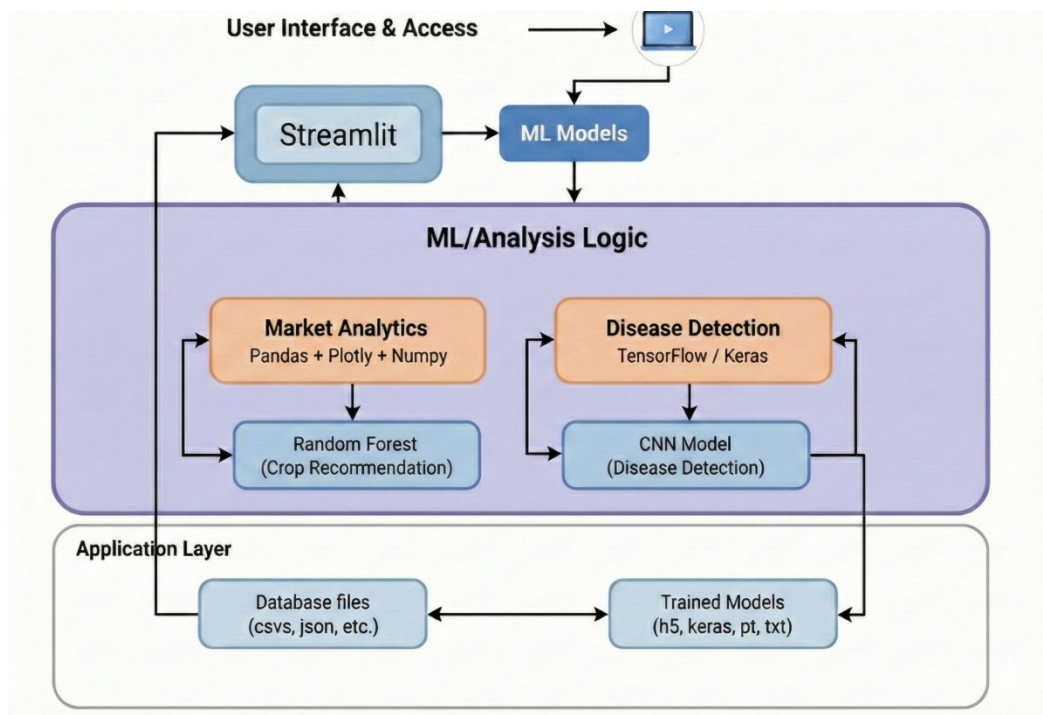


Figure 4.3 : System Architecture

The diagram represents a complete Python-based architecture designed to support agricultural decision-making. At the frontend, the system utilizes **Streamlit**, a lightweight and user-friendly framework that enables rapid development of interactive web interfaces without the complexity of traditional web technologies. This allows farmers and users to easily input soil values, upload leaf images, and view analytical outputs through a clean and responsive interface.

For data handling and computational tasks, the system incorporates powerful Python libraries such as **Pandas** and **NumPy**, which manage tabular data, perform numerical operations, and support preprocessing steps required for model execution. These libraries ensure efficient manipulation of soil datasets, image metadata, and market data used within the system.

The core intelligence of the platform is driven by machine learning models developed using **TensorFlow** and **Keras**. A **Random Forest model** is employed to generate crop recommendations based on soil parameters, while a **Convolutional Neural Network (CNN)** processes leaf images to diagnose plant diseases. Together, these models deliver accurate predictions and insights tailored to agricultural needs.

CHAPTER 5:

SYSTEM IMPLEMENTATION

The implementation phase involves translating the design into a functional system using the selected software and hardware components. As a prototype, the focus is on the successful training and deployment of the core AI/ML models and the setup of the information system.

Implementation involves dataset collection, preprocessing, model training, and deployment. The disease detection model can be trained on labeled leaf images using CNN architectures[8]. Crop recommendation uses features like pH, NPK, organic carbon, and moisture. The web interface collects inputs and displays outputs.

5.1 Implementation of Proposed Methodology

The proposed methodology follows a systematic approach consisting of data preparation, model development, and prediction. Initially, a soil–crop dataset containing parameters such as pH, NPK ratio, moisture content, and organic matter is collected and pre-processed using Python libraries like Pandas and NumPy to ensure accuracy and consistency. The refined dataset is then used to train a Random Forest classification model, which learns meaningful patterns and relationships between soil characteristics and suitable indigenous fruit crops. Once trained, the system accepts real-time soil input from the user and utilizes the model to generate an accurate crop recommendation. This methodology enables data-driven agricultural decision-making, helping farmers select the most suitable crop based on their soil profile.

1. Soil-Based Crop Suggestion Implementation

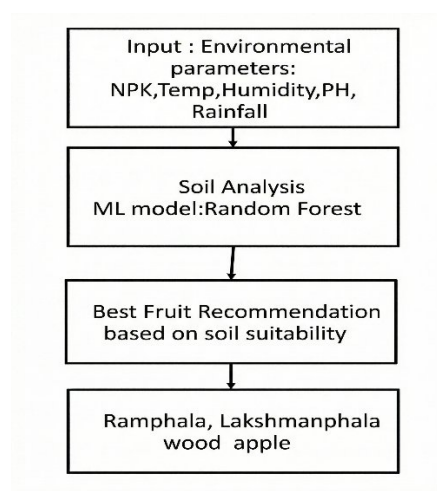


Figure 5.1 : Soil-Based Crop Suggestion Implementation

- **Data Preparation:** The process begins with sourcing an existing dataset that links diverse soil characteristics (pH, NPK values, organic matter, and moisture) to historically successful crops. The data is cleaned and pre-processed using **Pandas** and **NumPy**.
- **Model Training:** A **Random Forest** classification model from the **scikit-learn** library is trained on this prepared dataset. This model learns the complex relationship between soil inputs and optimal crop outputs.
- **Prediction:** When a farmer inputs their current soil data, the trained model compares these values against the patterns it has learned and predicts the most suitable indigenous fruit crop, thereby guiding planting decisions.

2. AI-Derived Disease Detection Implementation

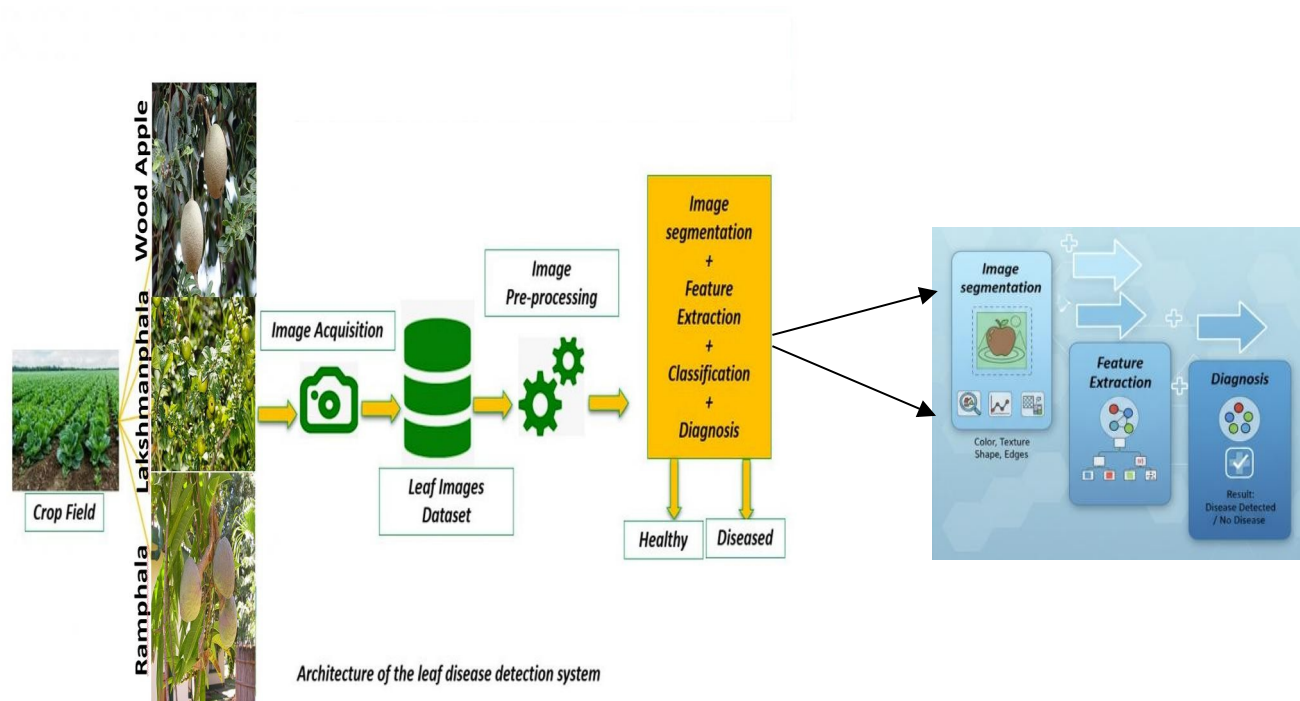


Figure 5.2 :Data Flow diagram of disease detection

- **Image Preprocessing Pipeline:** Pillow (PIL) opens and converts the uploaded leaf image, while Torchvision applies necessary transformations for standardization and input to the deep learning model[11]. OpenCV may be utilized for initial image manipulation or feature extraction[10][13].
- **Model Architecture:** A pre-trained **Convolutional Neural Network (CNN)**, such as a ResNet architecture (via Torchvision) , is loaded using the **PyTorch** framework.
- **Training & Classification:** The CNN is fine-tuned on a dataset of images showing the various diseases of the indigenous fruits (e.g., Anthracnose, Powdery Mildew). The model then classifies a new uploaded image, enabling swift and accurate diagnosis of plant diseases.

3. Marketing Strategy Implementation



Marketing & Growth Insights

Farmer Insights Consumer Insights



Insights for Cultivators

Estimated Market Price (India)

	Estimated Price	Notes
Ramphal a	₹90 - ₹250 / kg (Retail/Specialty)	Price varies significantly by market (e.g., Gujarat vs. Delhi) and quality. Mandi (wholesale) price approx. ₹9000-₹9900 / Quintal.
Lakshma nphala	₹300 - ₹800 / kg (Retail)	Very high price, often sold 'per piece'. Driven by health/nutraceutical demand. Dried leaves are also a valuable byproduct.
Wood Apple	₹50 - ₹150 / kg (Retail)	Lower per-fruit price but high yield per tree. Value is often added by processing into pulp, juice, and powder.

*Note: Prices are estimates based on available 2024-2025 online retail and mandi data. Actual prices will vary significantly by location, season, and quality.

Figure 5.3 :Marketing and Growth insight

- **Data Repository:** A central database is implemented to securely store all essential data, including the detailed cultivation requirements, nutritional/medicinal benefits, and market insights for Ramphala, Lakshmanphala, and Wood Apple.
- **Visualization Dashboard:** Tools like **Matplotlib/Seaborn** are used to generate dynamic graphs showing market trends, nutritional breakdowns, and crop value. A more comprehensive dashboard, potentially using **Power BI**, would be designed to give stakeholders and farmers strategic insights for effective consumer education and market promotion (e.g., Local Market Penetration, Digital Marketing, Consumer Engagement).

CHAPTER 6:**TEST CASES**

Use Case	Actor	Description	Pre-condition s	Post-conditions
UC1: View Fruit Information	Farmer/Consumer	The user selects a specific indigenous fruit (Ramphala, Lakshmanphala, or Wood Apple) to view its details.	Fruit details are stored in the DBMS.	Benefits, cultivation requirements, major diseases, and Nutritional and medicinal data are displayed.
UC2: Disease Diagnosis	Farmer	The farmer captures and uploads an image of a leaf showing symptoms to the AI model.	Internet connection is active; Leaf image is clear; AI model is trained and deployed.	Specific disease name is returned; Recommended treatment/management is displayed (e.g., Spray Carbendazim).

Use Case	Actor	Description	Pre-condition s	Post-conditions
UC3: Crop Recommendation	Farmer	The farmer inputs the result of their soil test (pH, NPK, etc.) and region to receive a suggestion for the optimal crop to cultivate.	Soil test results are available; System is functional.	Optimal indigenous crop(s) is/are suggested;

The system provides three main features for farmers and consumers. First, in the Crop Recommendation use case, the farmer enters soil test results such as pH and NPK along with their region, and the system analyzes these inputs to suggest the most suitable indigenous crop for cultivation. Second, in the Disease Diagnosis use case, the farmer uploads a clear image of a diseased leaf, and with an active internet connection and a trained AI model, the system identifies the specific disease and provides appropriate treatment or management advice. Third, through the View Fruit Information use case, the user selects an indigenous fruit like Ramphala, Lakshmanphala, or Wood Apple, and the system retrieves stored data to display its nutritional benefits, medicinal value, cultivation requirements, and major diseases. Together, these use cases help farmers make informed decisions, improve crop productivity, and understand indigenous fruit resources effectively.

6.1 Use Case Diagrams

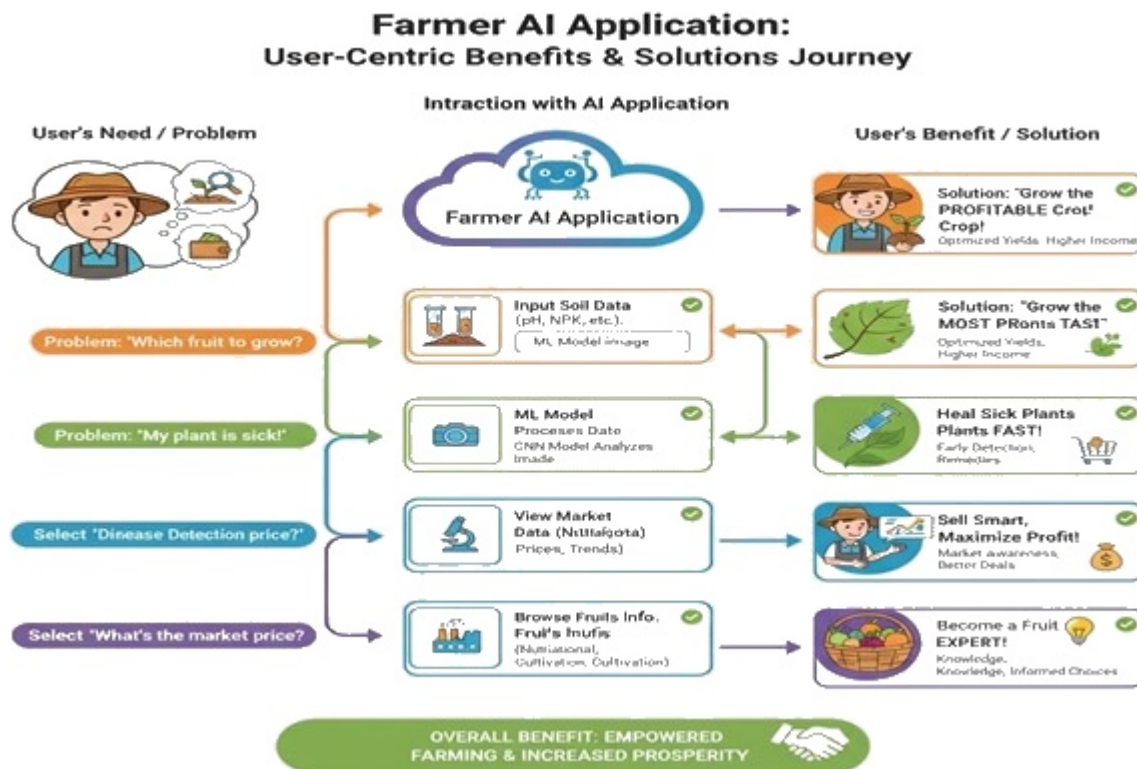


Figure 5.3 :Use Case Diagram

1. Soil Input Test: Enter typical soil values; expected: recommended crops match dataset label.
2. Disease Detection Test: Upload clear leaf image of known disease; expected: model predicts correct disease with high confidence.
3. Market Data Visualization: charts display correct trends.
4. End-to-End Test: Farmer uploads data and receives recommendation and disease alert.

The use case diagram represents the core interactions between the farmer and the system, illustrating how different functionalities are accessed and executed. The Soil Input Test evaluates the crop recommendation module by allowing the farmer to enter essential soil parameters such as pH, nitrogen, phosphorus, and potassium. The system processes these inputs using its trained dataset, and the expected theoretical outcome is that the recommended crops must accurately align with the ground-truth labels present in the model's data.

CHAPTER 7:**RESULTS****7.1 Implications drawn from the result :****Comparison of Machine Learning Models**

Model	How it Works (Simple)	Why it's Suitable (Pros)	Weaknesses (Cons)	Result (Accuracy)
Logistic Regression	Finds a line (or hyperplane) that separates classes based on probability.	Fast and simple to train. Works well when data is linearly separable. Great baseline model.	Cannot handle complex, non-linear data. Too simple for real-world patterns.	95.08%
K-Nearest Neighbors (KNN)	Stores all data; predicts by finding the <i>k</i> most similar data points and taking a vote.	Captures complex, non-linear relationships. Works well for clustered data.	Very slow on large datasets. Sensitive to noisy or irrelevant features.	99.45%
Decision Tree	Builds a tree of "if-then-else" rules based on feature values.	Easy to understand and visualize. Mimics human decision-making.	Overfits easily (memorizes training data). Small data changes can alter the tree.	100%
Random Forest	Combines many decision trees and averages their predictions.	Reduces overfitting. More accurate and stable than a single tree. Handles missing and categorical data well.	Hard to interpret ("black box"). Requires more computation and time.	100%
Gradient Boosting	Builds trees sequentially, each one correcting the mistakes of the previous tree.	Highly accurate and powerful. Learns from previous mistakes. Excellent for structured/tabular data.	Slow to train (sequential). Hard to interpret. Sensitive to noisy data.	100%

7.2 SNAPSHOTS

Include screenshots from your project here, such

**Fruit Information**

Detailed information on the featured indigenous fruits.

Ramphala (Bull's Heart / *Annona reticulata*)



Ramphala

Cultivation Factors

- **Climate:** Thrives in warm, humid, tropical, and subtropical climates.
- **Temperature:** Prefers temperatures between 20°C to 35°C. It is sensitive to frost, especially when young.
- **Sunlight:** Requires bright, direct sunlight (at least 6 hours).
- **Soil:** Flourishes in rich, well-draining loamy soil. It can tolerate various soil types, including shallow or sandy soil, but good drainage is crucial to prevent root rot.
- **Water:** Needs regular watering to keep the soil moist, but dislikes waterlogging.
- **Propagation:** Can be propagated by seeds or grafting. Grafted plants bear fruit faster.

Nutritional Value (per 100g)

- **Energy:** ~101 Kcal
- **Vitamin C:** ~19.2 mg (approx. 21-23% DV)
- **Vitamin B6:** ~0.221 mg (approx. 17% DV)
- **Potassium:** ~382 mg (approx. 13% DV)
- **Dietary Fiber:** ~2.4 g
- **Protein:** ~1.7 g
- **Also Contains:** Magnesium, Calcium, and antioxidants (polyphenols, flavonoids).

Medicinal & Economic Uses

- **Culinary:** Eaten fresh, or used in juices, smoothies, desserts, and ice cream.
- **Immunity:** High Vitamin C content helps boost the immune system.
- **Digestive Health:** Fiber aids in digestion and helps regulate bowel movements.
- **Heart Health:** Potassium helps regulate blood pressure, and fiber can help lower cholesterol.
- **Skin Health:** Vitamin C and antioxidants help protect skin and boost collagen production.
- **Traditional Medicine:** Unripe fruit used for diarrhea/dysentery; leaves and bark used for various ailments like ulcers, boils, and fever.

Figure 7.1 Information about Ramphala

Ramphala, also known as Bull's Heart (*Annona reticulata*), is an indigenous tropical fruit valued for its sweet flavor and rich nutrition. It thrives in warm, well-drained soils with regular watering and sunlight. The fruit is packed with vitamins C and B6, potassium, fiber, and antioxidants that support immunity, digestion, and heart health. Beyond its culinary uses in juices and desserts, it holds importance in traditional medicine for treating ailments like diarrhea and fever, making it both nutritionally and economically significant.

Lakshmanphala (Soursop / *Annona muricata*)



Lakshmanphala

Cultivation Factors

- **Climate:** Requires a wet, tropical or subtropical climate with high humidity.
- **Temperature:** Prefers warm temperatures (25°C - 30°C). It is highly sensitive to frost and cannot withstand temperatures below 5°C.
- **Sunlight:** Needs full, direct sun exposure (at least 8 hours).
- **Soil:** Prefers deep, well-drained, sandy loam soil rich in organic matter.
- **Soil pH:** Ideal pH is slightly acidic, between 5.5 to 6.5.
- **Water:** Needs consistently moist soil; avoid waterlogging, which causes root rot.
- **Propagation:** Grown from seeds or grafting. Grafted trees bear fruit earlier (3-5 years).

Nutritional Value (per 100g)

- **Energy:** ~66 Kcal
- **Vitamin C:** ~21 mg (approx. 23% DV). (Excellent source overall).
- **Potassium:** ~278 mg (approx. 8% DV).
- **Dietary Fiber:** ~3.3 g (approx. 13% DV). (Very high fiber content).
- **Protein:** ~1.0 g
- **Also Contains:** Good source of B vitamins (B1, B2, B3, B6, Folate), Magnesium, and Copper.

Medicinal & Economic Uses

- **Culinary:** Very popular for juices, smoothies, sorbets, and desserts. Also eaten fresh.
- **Economic:** High market demand both locally and internationally, increasing profitability.
- **Immunity:** Excellent source of Vitamin C, a known immune booster.
- **Digestive Health:** Very high fiber content promotes regular bowel movements.
- **Antioxidant/Anti-inflammatory:** Rich in antioxidants (phenolic compounds, acetogenins).
- **Traditional Medicine:** Leaves are famously brewed into a tea used to treat a variety of ailments, including inflammation, pain, and infections.

Figure 7.2: Information about Lakshmanphala

Lakshmanphala, commonly known as Burmese grape, is a juicy indigenous fruit grown in humid tropical regions. It contains vitamin C, iron, and antioxidants that boost immunity and aid digestion. Besides its pleasant taste, the fruit has medicinal uses for treating anemia and gastrointestinal issues, making it valuable for both nutrition and rural livelihoods.

Wood Apple (Bael / *Limonia acidissima*)



Wood Apple

Cultivation Factors

- **Climate:** Extremely hardy; thrives in seasonally dry tropical or subtropical monsoon climates. Prefers a distinct dry season.
- **Temperature:** Prefers mean annual temperatures from 20°C to 35°C.
- **Sunlight:** Grows best in full sun.
- **Soil:** Not particular. Tolerates a very wide range of soils, including poor, sandy, or light soils. Best in well-drained sandy or deep loam soils.
- **Tolerance:** Highly drought-tolerant once established.
- **Propagation:** Commonly grown from seeds, but budded plants bear fruit much earlier.

Nutritional Value (per 100g)

- **Energy:** ~134-137 Kcal
- **Carbohydrates:** ~18-31.8 g
- **Protein:** ~1.8-7.0 g
- **Fiber:** ~2.9-5.0 g
- **Vitamins:** Rich in B vitamins, especially Riboflavin (B2) and Thiamine (B1). Contains Vitamin C and Beta-carotene (Vitamin A precursor).
- **Minerals:** Good source of Calcium (85-130 mg) and Potassium (600 mg).

Medicinal & Economic Uses

- **Culinary:** Pulp is famously processed into a refreshing juice (sherbet), jams (murabba), and chutneys.
- **Digestive Health:** Highly valued in Ayurveda. The *unripe* fruit is used to treat diarrhea and dysentery. The *ripe* fruit pulp is an excellent laxative.
- **Medicinal:** Considered a natural antioxidant and cardio-protective fruit. Used in traditional medicine for respiratory issues, to purify blood, and to manage blood sugar.
- **Other Uses:** The hard shell is used for crafts, and gum from the trunk is used in inks and dyes.
- **Cosmetic:** Antimicrobial properties help treat scalp infections and dandruff; antioxidants aid skin health.

Figure 7.3: Information about Wood Apple

Wood Apple is a hardy indigenous fruit known for its hard shell and aromatic, tangy pulp. It thrives in dry tropical regions and is rich in vitamin C, calcium, and fiber. The fruit aids digestion, boosts immunity, and is widely used in traditional medicine and beverages, making it both nutritional and economical[6]. .

Soil-Based Crop Suggestion Implementation



Smart Crop Recommendation & Similar Crop Finder

Enter Environmental Parameters

Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature (°C)

Humidity (%)

pH Value

Rainfall (mm)

Recommended Crop: Lakshmanphal

Other Crops with Similar Conditions

Lakshmanphal – Suitability: 39.78%

Wood apple – Suitability: 0.55%

Ramphal – Suitability: 0.00%



Figure 7.4 : Soil-Based Crop Suggestion Implementation

The Smart Crop Recommendation and Similar Crop Finder system analyzes environmental parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall to suggest the most suitable indigenous fruitcrop for cultivation.

Based on the provided data, the system recommends **Lakshmanphala** as the most suitable crop, with **Wood Apple** and **Ramphal** identified as alternative crops with similar growing conditions. This helps farmers make data-driven decisions to maximize yield and ensure sustainable crop selection.

The screenshot displays the 'Simple Dataset Uploader (Disease Name Only)' interface. At the top, three status messages are shown: a blue box indicating 'Processing save for disease class: 'Phoma_Leaf_Spot'...', a green box stating '2 new image(s) saved to: my_plant_dataset_disease_only\Phoma_Leaf_Spot', and a yellow box noting '15 image(s) skipped (already existed)'. Below these is the title 'Simple Dataset Uploader (Disease Name Only)' with a document icon. A note says 'Add images. Images will be saved into folders named after the Disease Name.' and a reminder box states 'Remember: Re-run 2_train_model.py (update DATASET_PATH if needed) after adding enough images!'. The workflow is divided into two sections: '1. Define Disease Label' and '2. Upload Images'. In the first section, a text input field contains 'Phoma Leaf Spot' with a prompt 'Enter Disease Name (e.g., Leaf_Spot, Healthy) - THIS DEFINES THE FOLDER'. The second section, '2. Upload Images', includes a 'Choose images...' prompt, a file upload area with a cloud icon and text 'Drag and drop files here' and 'Limit 200MB per file • JPG, JPEG, PNG', a 'Browse files' button, and a file list showing 'images.jpg 7.3KB' with a close icon.

Processing save for disease class: 'Phoma_Leaf_Spot'...

2 new image(s) saved to: my_plant_dataset_disease_only\Phoma_Leaf_Spot

15 image(s) skipped (already existed).

Simple Dataset Uploader (Disease Name Only)

Add images. Images will be saved into folders named after the **Disease Name**.

Remember: Re-run 2_train_model.py (update DATASET_PATH if needed) after adding enough images!

1. Define Disease Label

Enter Disease Name (e.g., Leaf_Spot, Healthy) - THIS DEFINES THE FOLDER

Phoma Leaf Spot

2. Upload Images

Choose images...

Drag and drop files here
Limit 200MB per file • JPG, JPEG, PNG

Browse files

images.jpg 7.3KB

Figure 7.5 Disease Dataset Uploader

The **Simple Dataset Uploader** allows users to upload and organize plant disease images by defining a disease label. In this example, images for the disease “**Phoma Leaf Spot**” are being saved into a dedicated folder. The system confirms new uploads, skips existing files, and provides reminders to retrain the model after adding sufficient images. This tool simplifies dataset creation for machine learning-based plant disease detection.

The system intelligently manages files by confirming newly added images and skipping duplicates, ensuring the dataset remains clean and well-organized. This minimizes redundancy and reduces manual effort during data preparation. Additionally, the tool provides prompts reminding the user to retrain the model once an adequate number of new images have been added, ensuring that the AI system stays updated and improves in accuracy over time.

Overall, the Simple Dataset Uploader simplifies and accelerates the dataset creation process, making it more efficient for users who want to train or enhance plant disease classification models.

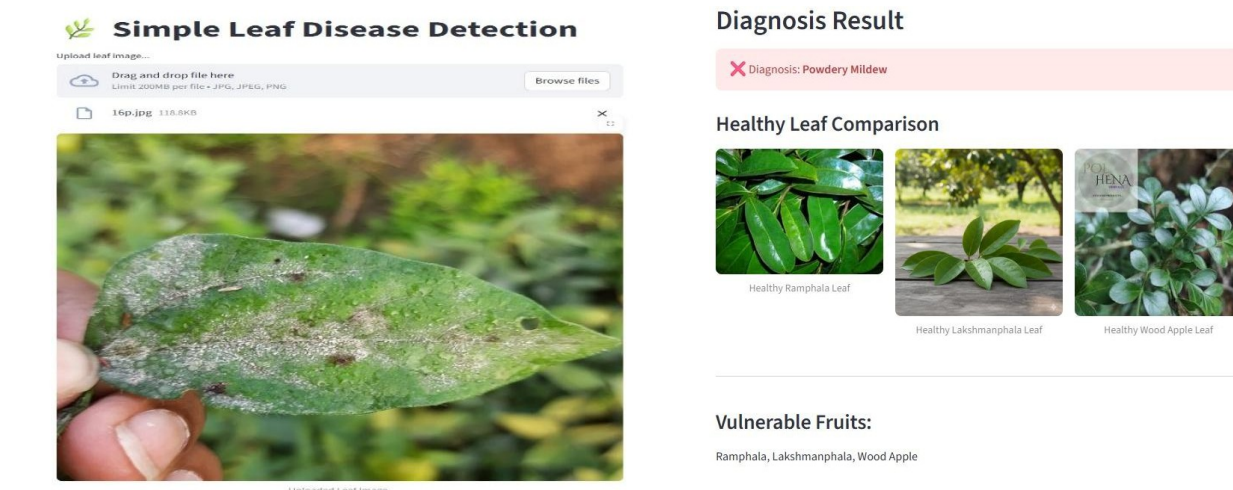


Figure 7. 6 : Leaf Disease Detection

The image shows a leaf disease detection result. The uploaded leaf is covered in a noticeable white, powdery substance.

The application, "Simple Leaf Disease Detection," provided a diagnosis of Powdery Mildew. This is a common fungal disease that affects many plants.

The screen also offers comparisons with images of healthy Ramphala, Lakshmanphala, and Wood Apple leaves.

Marketing & Growth Insights

Farmer Insights Consumer Insights

Insights for Cultivators

Estimated Market Price (India)

	Estimated Price	Notes
Ramphala	₹90 - ₹250 / kg (Retail/Specialty)	Price varies significantly by market (e.g., Gujarat vs. Delhi) and quality. Mandi (wholesale) price approx. ₹9000-₹9900 / Quintal.
Lakshmanphala	₹300 - ₹800 / kg (Retail)	Very high price, often sold 'per piece'. Driven by health/nutraceutical demand. Dried leaves are also a valuable byproduct.
Wood Apple	₹50 - ₹150 / kg (Retail)	Lower per-fruit price but high yield per tree. Value is often added by processing into pulp, juice, and powder.

*Note: Prices are estimates based on available 2024-2025 online retail and mandi data. Actual prices will vary significantly by location, season, and quality.



Marketing & Growth Insights

Data sourced from 'Indian Fruit Research: Uses and Potential.pdf'

Farmer Insights

Consumer Insights



Insights for Consumers

Health Benefits

Ramphala

- Traditionally used to treat diarrhea, dysentery, and ulcers.
- Possesses significant antioxidant, anti-inflammatory, and antimicrobial properties.

Lakshmanphala

- Rich in Vitamin C, which helps boost immunity.
- High fiber content aids in digestion and maintains gastrointestinal health.
- Known for anti-inflammatory and antioxidant properties that combat cellular damage.
- Widely studied for its acetogenin compounds and their potential anticancer effects.

Wood Apple

- Highly valued in traditional medicine; ripe fruit acts as a laxative, while unripe fruit helps treat diarrhea and dysentery.
- Manages digestive disorders and is known to have liver-protecting properties.
- Exhibits antimicrobial and antiviral activities.

Figure 7.7 : Market and Growth Insight

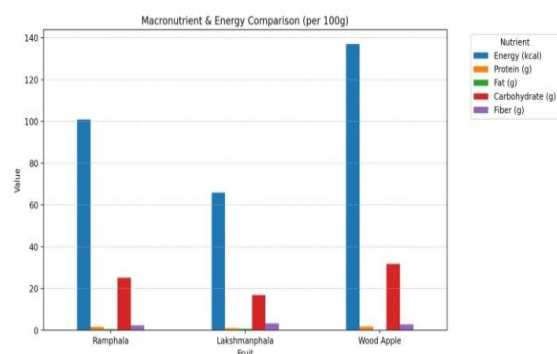
These insights help farmers understand market trends and potential returns while guiding consumers about the nutritional and medicinal benefits of native fruits. Promoting indigenous crops like Ramphala, Lakshmanphala, and Wood Apple can enhance both farmer income and public health awareness.



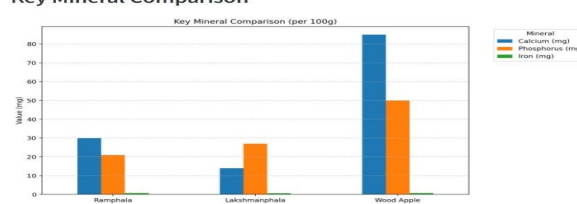
Nutritional Comparison (per 100g)

Data from Tables 1, 2, and 3 in the research PDF.

Macronutrients & Energy



Key Mineral Comparison



Vitamin C Comparison

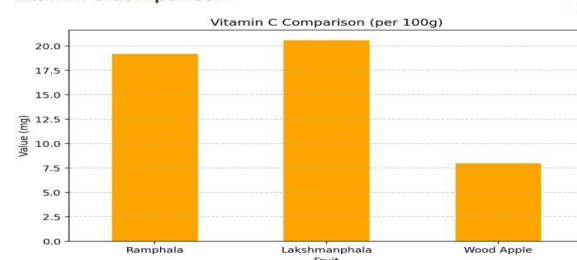


Figure 7.8: Nutritional Comparison

The nutritional comparison of Ramphala, Lakshmanphala, and Wood Apple (per 100g) shows notable differences in their nutrient composition. Wood Apple provides the highest energy and carbohydrate content, making it a rich energy source, while Ramphala and Wood Apple also offer a good amount of dietary fiber, beneficial for digestion. Protein and fat contents are low across all three fruits. In terms of minerals, Wood Apple stands out with the highest levels of calcium and phosphorus, followed by Ramphala and Lakshmanphala, indicating its mineral richness. The Vitamin C comparison reveals that Lakshmanphala contains the highest amount of Vitamin C, closely followed by Ramphala, whereas Wood Apple has comparatively lower Vitamin C content. Overall, Wood Apple is the most nutrient-dense fruit in terms of energy and minerals, while Lakshmanphala and Ramphala are excellent sources of Vitamin C, making all three fruits valuable for a balanced diet and good health.

CONCLUSION WITH FUTURE ENHANCEMENTS

Conclusion

The "Smart Cultivation and Marketing of Indigenous Fruit Crops" project successfully outlines a model to empower farmers through an integrated technological solution. By achieving its primary goals—enhancing cultivation through a data-driven crop suggestion module, protecting yield via an AI disease detection system, and ensuring profitability through strategic market promotion—the project provides a sustainable and efficient framework for popularizing and profiting from valuable indigenous fruits. This prototype demonstrates the powerful role that AI and data science can play in transforming the agricultural sector, particularly for specialized, high-value crops.

Future Enhancements

Potential future work can expand the scope of the project:

- **Mobile Application Development:** Developing a dedicated, fully functional mobile app (using Android/iOS) for farmers to easily input data and upload leaf images in the field.
- **Real-Time Market Pricing:** Integrating live APIs to fetch and display the current market price for the suggested fruits, improving negotiation power for farmers.
- **Irrigation/Fertilizer Recommendation:** Expanding the ML model to include precise, personalized recommendations for irrigation scheduling and integrated nutrient management (INM) based on moisture and NPK data.
- **Pest and Insect Detection:** Extending the AI vision model to not only detect diseases but also identify common pests and insects affecting the indigenous fruit crops.
- **Multi-lingual Support:** Implementing the interface and information in local regional languages to ensure maximum accessibility and utility for farmers.

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